Experiment -4

Python Program To Crypt-Arithmetic Problem

Code:

```
from itertools import permutations

letters = "SENDMORY"

for p in permutations(range(10), 8):
    d = dict(zip(letters, p))

    if d['S'] == 0 or d['M'] == 0:
        continue

    SEND = (d['S']*1000 + d['E']*100 +
            d['N']*10 + d['D'])

    MORE = (d['M']*1000 + d['O']*100 +
            d['R']*10 + d['E'])

    MONEY = (d['M']*10000 + d['O']*1000 +
             d['N']*100 + d['E']*10 + d['Y'])

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        break
```

```python
from itertools import permutations

letters = "SENDMORY"

for p in permutations(range(10), 8):
    d = dict(zip(letters, p))

    if d['S'] == 0 or d['M'] == 0:
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    SEND = (d['S']*1000 + d['E']*100 +
            d['N']*10 + d['D'])

    MORE = (d['M']*1000 + d['O']*100 +
            d['R']*10 + d['E'])

    MONEY = (d['M']*10000 + d['O']*1000 +
             d['N']*100 + d['E']*10 + d['Y'])

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        break
```

```
SEND = 9567
MORE = 1085
MONEY = 10652


...Program finished with exit code 0
Press ENTER to exit console.
```